

5 Qs

$$4 \times 5 \text{ marks} = 20$$

$$1 \times 10 \text{ marks} = 10$$

30

Total marks = 30

Origins of Quantum Theory

1. Blackbody radiation

$$E = h\nu$$

2. Photoelectric effect

$$KE = h\nu - \phi$$

3. Spectral lines of H-atom

$$= h\nu - h\nu_0$$

Rydberg's formula

$$\frac{1}{\lambda} = \tilde{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad n_2 > n_1$$

Lyman

$\frac{n_1}{1}$

Balmer

$\frac{n_2}{2}$

Paschen

$\frac{n_2}{3}$

Brackett

$\frac{n_2}{4}$

4. Bohr's Theory

Radius

$$r = \frac{n^2}{Z} a_0$$

Atomic Units

$$a \equiv \text{bohr} \\ (\text{bohr}) = 0.529 \text{ \AA}$$

Energy

$$E = - \frac{Z^2}{2n^2} E_h$$

$$E_h = 2 \times 13.6 \text{ eV} \\ (\text{hartree})$$

- de Broglie's wave-particle duality
- Heisenberg's Uncertainty Principle

Principles of QM

TISE

$$\hat{H}\psi = E\psi$$

Eigenvalue Eq.

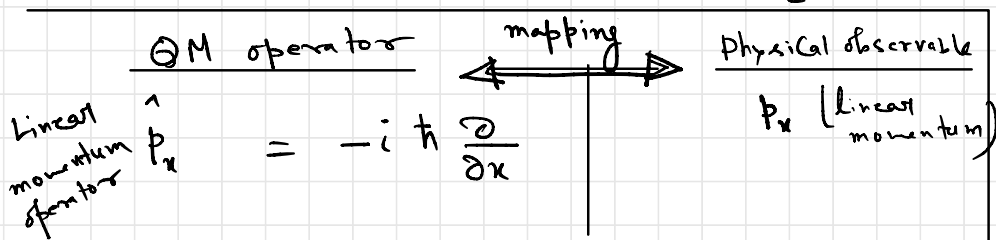
$$\hat{O}f(x) = Of(x)$$

Conditions
for well-behaved
Wavefunction

1. Continuous
2. finite
3. single-valued
4. Slope continuous

Hamiltonian operator $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$

$$\hbar = \frac{h}{2\pi}$$



Probability density $P(x) = |\psi|^2 = \psi^* \psi$

Given: $\psi = e^{-ikx}$

therefore, $\psi^* = e^{+ikx}$

$$P(x) = \psi^* \psi = e^{ikx} \cdot e^{-ikx} = e^0 = 1$$

Probability within a space/region $\int_0^a \psi^* \psi dx = ?$

SHOWING $\hat{H}$ OPERATOR HERMITIAN FOR A GIVEN FUNCTION

$\hat{H}$: hermitian & linear

(Hamiltonian operator)

→ will produce real eigenvalue

$$\Rightarrow \left[\underbrace{\hat{H}}_{\text{operator}} f(x) \right] \quad \text{function} \quad \left[\sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \right]$$

$$\Rightarrow \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \underbrace{V(x)}_{\substack{\parallel \\ 0}} \right] \left[\sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \right]$$

$$\Rightarrow -\frac{\hbar^2}{2m} \times \sqrt{\frac{2}{a}} \left(-\sin \frac{n\pi x}{a} \right) \left(\frac{n\pi}{a} \right)^2 \quad \left| \quad \begin{aligned} f(x) &= \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \\ \hbar &= \frac{h}{2\pi} \end{aligned} \right.$$

$$\Rightarrow \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2 \underbrace{\sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}}_{f(x)}$$

$$\Rightarrow \frac{\hbar^2 n^2 \pi^2}{8\pi^2 m a^2} f(x)$$

$$\Rightarrow \frac{n^2 \hbar^2}{8\pi^2 m a^2} f(x)$$

real eigenvalue → hence, hermitian

Free - particle

Particle in 1-D box

1. $-\infty < x < +\infty$

2. $\hat{H}\Psi = E\Psi$

3. $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

4. $\Psi_k(x) = e^{ikx} + e^{-ikx}$
 $\equiv C \cos kx + D \sin kx$
 here, k is continuous!

5. $E = \frac{\hbar^2 k^2}{2m}$

E is continuous!
 NOT QUANTIZED

For ex: $k = 0$

$$E = 0$$

Free particle can be at rest!

1. Particle is confined over a space in region (e.g., 0 to a)

2. $\hat{H}\Psi = E\Psi$

3. $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

4. $\Psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$

$n = 1, 2, 3, 4, \dots$

5. $E = \frac{n^2 \hbar^2}{8ma^2}$

$n = 1, 2, 3, \dots$

E is QUANTIZED!

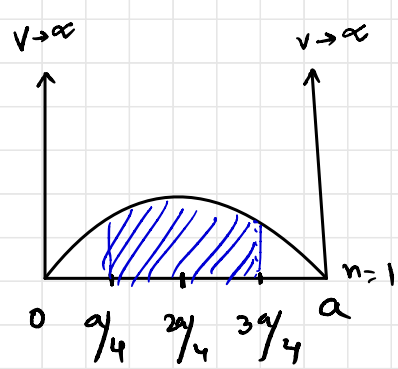
For ex $n = 1$

$$E = \frac{\hbar^2}{8ma^2}$$

At lowest-energy state (GROUND-STATE), particle has certain amount of energy! Never at rest!

* here, ONLY Translational motion

What is the probability of finding the particle in region $\frac{a}{4} \leq x \leq \frac{3a}{4}$ for a particle in 1D box?



Ans

$$\psi = \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} \quad \text{for } n=1$$

$$P = \int_{a/4}^{3a/4} \psi^* \psi \, dx = \int_{a/4}^{3a/4} \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} \times \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} \, dx$$

$$= \int_{a/4}^{3a/4} \frac{2}{a} \sin^2 \frac{\pi x}{a} \, dx$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$= \frac{2}{a} \left[\int_{a/4}^{3a/4} \frac{1}{2} (1 - \cos \frac{2\pi x}{a}) \, dx \right]$$

$$= \frac{1}{a} \times \frac{2}{a} \left[\int_{a/4}^{3a/4} dx - \int_{a/4}^{3a/4} \cos \frac{2\pi x}{a} \, dx \right]$$

$$= \frac{1}{a} \left[a \left| \begin{matrix} 3a/4 \\ a/4 \end{matrix} \right| - \frac{\sin \frac{2\pi x}{a}}{2\pi/a} \left| \begin{matrix} 3a/4 \\ a/4 \end{matrix} \right| \right]$$

$$= \frac{1}{a} \left[\left\{ \frac{3a}{4} - \frac{a}{4} \right\} - \frac{a}{2\pi} \left\{ \sin \frac{\cancel{2}\pi \times 3\cancel{a}}{\cancel{a} \times \cancel{4}} - \sin \frac{\cancel{2}\pi \times \cancel{a}}{\cancel{a} \times \cancel{4}} \right\} \right]$$

$$= \frac{1}{a} \times \frac{a}{2} - \frac{1}{2\pi} \{-1 - 1\}$$

$$= \frac{1}{2} + \frac{1}{\pi}$$

ORTHONORMALITY CONDITION

$$\int_{\text{all space}} \psi_m^* \psi_n dx = 1 \quad \text{for } m = 1$$

NORMALIZED condition

$$= 0 \quad \text{for } m \neq n$$

ORTHOGONAL condition